ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Arithmetic Progressions** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 40	First find $d = 3$
2	**(a)** 24	$n = \frac{l-a}{d} + 1$
3	**(c)** 1665	AP of two-digit multiples
4	**(c)** 623	Use S_n
5	**(b)** 23	Middle term $= \frac{a+l}{2}$
6	**(a)** 4	$a = a_p - (p-1)d$
7	**(a)** 52	First find $d = 6$
8	**(c)** 31	$n = \frac{l-a}{d} + 1$
9	**(c)** 1188	AP of two-digit multiples
10	**(a)** 640	Use S_n

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$a_n = a + (n-1)d$: $127 = 7 + (n-1)(6)$ (1); $n - 1 = 20$ (1); $n = 21$, so it is the 21st term (1).	1+1+1
12	$a = 14$, $d = 6$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{18} = \frac{18}{2}[28 + 17(6)] = 1170$ (1).	1+1+1
13	First term 4, last term 96 (1); $n = 24$, sum $= 1200$ (1).	1+1
14	$a_n = a + (n-1)d$: $79 = 11 + (n-1)(4)$ (1); $n - 1 = 17$ (1); $n = 18$, so it is the 18th term (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$a + 2d = 19$, $a + 7d = 54$ (2); subtracting, $d = 7$, $a = 5$ (2); $a_{15} = 103$ (1).	2+2+1
16	$a_1 = 5$, $a_{20} = 62$ (1); $d = 3$ (1); $S_{20} = \frac{20}{2}(5 + 62) = 670$ (1).	1+1+1
17	$S_n = \frac{n}{2}[2a + (n-1)d]$: $185 = \frac{10}{2}[10 + 9d]$ (1); $d = 3$ (1); $a_{10} = 32$ (1).	1+1+1
	OR: $a + d = 2$, $a + 6d = 32$ (2); subtracting, $d = 6$, $a = -4$ (2); $a_{12} = 62$ (1).	2+2+1
18	$a_1 = 4$, $a_{15} = 74$ (1); $d = 5$ (1); $S_{15} = \frac{15}{2}(4 + 74) = 585$ (1).	1+1+1
	OR: $S_n = \frac{n}{2}[2a + (n-1)d]$: $360 = \frac{12}{2}[16 + 11d]$ (1); $d = 4$ (1); $a_{12} = 52$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$S_n = \frac{n}{2}[2(54) + (n-1)(-4)] = 360$ (1); form quadratic in n (2); $n = 10$ (the other root is rejected) (2).	1+2+2
	OR: AP with $a = 45$, $l = 25$, $n = 11$ (1); $S_n = \frac{11}{2}(45 + 25) = 385$ cm (2); middle term $= \frac{45+25}{2} = 35$ cm (2).	1+2+2
20	$S_n = \frac{n}{2}[2(12) + (n-1)(6)] = 714$ (1); form quadratic in n (2); $n = 14$ (the other root is rejected) (2).	1+2+2
	OR: AP with $a = 60$, $l = 30$, $n = 13$ (1); $S_n = \frac{13}{2}(60 + 30) = 585$ cm (2); middle term $= \frac{60+30}{2} = 45$ cm (2).	1+2+2

Mistakes to avoid:

- Using $a_n = a + nd$ instead of $a + (n-1)d$.
- Writing $S_n = \frac{n}{2}[a + (n-1)d]$; the bracket needs $2a$.
- Taking d as $a_1 - a_2$; it is $a_2 - a_1$ (a decreasing AP has negative d).
- Using S_n when only a_n is asked (or the other way round) in word problems.
- Not rejecting a negative or fractional value of n obtained from a quadratic in n.
- Calling a list an AP after checking only the first two differences.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Arithmetic Progressions** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 28	Middle term $= \frac{a+l}{2}$
2	**(c)** 10	$a = a_p - (p-1)d$
3	**(c)** 39	First find $d = 2$
4	**(b)** 17	$n = \frac{l-a}{d} + 1$
5	**(d)** 945	AP of two-digit multiples
6	**(c)** 162	Use S_n
7	**(c)** 30	Middle term $= \frac{a+l}{2}$
8	**(d)** 7	$a = a_p - (p-1)d$
9	**(d)** 48	First find $d = 5$
10	**(a)** 26	$n = \frac{l-a}{d} + 1$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$a = -12$, $d = 4$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{26} = \frac{26}{2}[-24 + 25(4)] = 988$ (1).	1+1+1
12	First term 15, last term 195 (1); $n = 37$, sum $= 3885$ (1).	1+1
13	$a_n = a + (n-1)d$: $62 = -3 + (n-1)(5)$ (1); $n - 1 = 13$ (1); $n = 14$, so it is the 14th term (1).	1+1+1
14	$a = 6$, $d = \frac{3}{2}$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{20} = \frac{20}{2}[12 + 19(\frac{3}{2})] = 405$ (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$a + 3d = 3$, $a + 8d = -12$ (2); subtracting, $d = -3$, $a = 12$ (2); $a_{14} = -27$ (1).	2+2+1
16	$a_1 = 11$, $a_{12} = 55$ (1); $d = 4$ (1); $S_{12} = \frac{12}{2}(11 + 55) = 396$ (1).	1+1+1
17	$S_n = \frac{n}{2}[2a + (n-1)d]$: $660 = \frac{15}{2}[4 + 14d]$ (1); $d = 6$ (1); $a_{15} = 86$ (1).	1+1+1
	OR: $a + 2d = 18$, $a + 5d = 33$ (2); subtracting, $d = 5$, $a = 8$ (2); $a_{20} = 103$ (1).	2+2+1
18	$a_1 = 11$, $a_{25} = 59$ (1); $d = 2$ (1); $S_{25} = \frac{25}{2}(11 + 59) = 875$ (1).	1+1+1
	OR: $S_n = \frac{n}{2}[2a + (n-1)d]$: $600 = \frac{20}{2}[22 + 19d]$ (1); $d = 2$ (1); $a_{20} = 49$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$S_n = \frac{n}{2}[2(-10) + (n-1)(4)] = 144$ (1); form quadratic in n (2); $n = 12$ (the other root is rejected) (2).	1+2+2
	OR: AP with $a = 50$, $l = 20$, $n = 7$ (1); $S_n = \frac{7}{2}(50 + 20) = 245$ cm (2); middle term $= \frac{50+20}{2} = 35$ cm (2).	1+2+2
20	$S_n = \frac{n}{2}[2(36) + (n-1)(-3)] = 231$ (1); form quadratic in n (2); $n = 11$ (the other root is rejected) (2).	1+2+2
	OR: AP with $a = 80$, $l = 40$, $n = 9$ (1); $S_n = \frac{9}{2}(80 + 40) = 540$ cm (2); middle term $= \frac{80+40}{2} = 60$ cm (2).	1+2+2

Mistakes to avoid:

- Using $a_n = a + nd$ instead of $a + (n-1)d$.
- Writing $S_n = \frac{n}{2}[a + (n-1)d]$; the bracket needs $2a$.
- Taking d as $a_1 - a_2$; it is $a_2 - a_1$ (a decreasing AP has negative d).
- Using S_n when only a_n is asked (or the other way round) in word problems.
- Not rejecting a negative or fractional value of n obtained from a quadratic in n.
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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Arithmetic Progressions** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 810	AP of two-digit multiples
2	**(b)** 984	Use S_n
3	**(a)** 45	Middle term $= \frac{a+l}{2}$
4	**(c)** 1	$a = a_p - (p-1)d$
5	**(b)** 61	First find $d = 4$
6	**(d)** 19	$n = \frac{l-a}{d} + 1$
7	**(a)** 728	AP of two-digit multiples
8	**(d)** 517	Use S_n
9	**(a)** 51	Middle term $= \frac{a+l}{2}$
10	**(d)** 13	$a = a_p - (p-1)d$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	First term 6, last term 144 (1); $n = 24$, sum $= 1800$ (1).	1+1
12	$a_n = a + (n-1)d$: $8 = 30 + (n-1)(-2)$ (1); $n - 1 = 11$ (1); $n = 12$, so it is the 12th term (1).	1+1+1
13	$a = 27$, $d = -2$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{15} = \frac{15}{2}[54 + 14(-2)] = 195$ (1).	1+1+1
14	First term 21, last term 294 (1); $n = 40$, sum $= 6300$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$a + 4d = 38$, $a + 9d = 83$ (2); subtracting, $d = 9$, $a = 2$ (2); $a_{13} = 110$ (1).	2+2+1
16	$a_1 = 2$, $a_{10} = 56$ (1); $d = 6$ (1); $S_{10} = \frac{10}{2}(2 + 56) = 290$ (1).	1+1+1
17	$S_n = \frac{n}{2}[2a + (n-1)d]$: $172 = \frac{8}{2}[8 + 7d]$ (1); $d = 5$ (1); $a_8 = 39$ (1).	1+1+1
	OR: $a + d = 19$, $a + 10d = 55$ (2); subtracting, $d = 4$, $a = 15$ (2); $a_{18} = 83$ (1).	2+2+1
18	$a_1 = 10$, $a_{16} = 115$ (1); $d = 7$ (1); $S_{16} = \frac{16}{2}(10 + 115) = 1000$ (1).	1+1+1
	OR: $S_n = \frac{n}{2}[2a + (n-1)d]$: $472 = \frac{16}{2}[14 + 15d]$ (1); $d = 3$ (1); $a_{16} = 52$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$S_n = \frac{n}{2}[2(6) + (n-1)(5)] = 696$ (1); form quadratic in n (2); $n = 16$ (the other root is rejected) (2).	1+2+2
	OR: AP with $a = 75$, $l = 35$, $n = 17$ (1); $S_n = \frac{17}{2}(75 + 35) = 935$ cm (2); middle term $= \frac{75+35}{2} = 55$ cm (2).	1+2+2
20	$S_n = \frac{n}{2}[2(60) + (n-1)(-5)] = 360$ (1); form quadratic in n (2); $n = 9$ (the other root is rejected) (2).	1+2+2
	OR: AP with $a = 66$, $l = 30$, $n = 19$ (1); $S_n = \frac{19}{2}(66 + 30) = 912$ cm (2); middle term $= \frac{66+30}{2} = 48$ cm (2).	1+2+2

Mistakes to avoid:

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